

ADAM ZAATAR

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EDUCATION

Bowdoin College

Brunswick, ME

B.A., Computer Science and Economics, Minor in German

Expected May 2027 · GPA 3.53

Relevant coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Computer Systems, Econometrics, Artificial Intelligence

Learning Assistant, CSCI 1101 (Intro to CS), Bowdoin CS Department — Fall 2026

PROJECTS

Technical Bid Review

Python, FastAPI, PostgreSQL, OpenAI API

- Built an evidence-grounded, tool-using LLM agent that evaluates technical vendor proposals against an owner specification across five compliance states, scoring 10/10 classification accuracy with 0 false-compliance errors on a frozen, unseen synthetic holdout run once after the evaluator was frozen.
- Constrained the agent to two evidence tools over untrusted documents, with no shell, web, arbitrary SQL, or cross-vendor access, and kept authority in deterministic code: retrieval at 100% Recall@5, 12/12 expected-evidence provenance overlap on that holdout, citation and unit-aware numeric validation, immutable model proposals, and a separate human accept/override path.

Durable Runner

TypeScript, Node.js, PostgreSQL, Fastify, React

- Built a PostgreSQL-backed durable execution runtime for long-running, side-effecting workloads: independent worker processes claim work with FOR UPDATE SKIP LOCKED, hold renewable leases, and are fenced by a monotonic generation token, so a frozen worker that resumes late cannot overwrite reassigned work, while a stable logical-effect key lets its replacement reuse an effect it already performed rather than repeating it.
- Benchmarked 3,277 tasks/sec median across 8 workers, with 0 duplicate grants over 200,266 concurrent ownership claims, 0 accepted stale writes over 4,750 stale-worker mutations, and 0 duplicate logical effects over 45,000 executions; failure paths proven with SIGSTOP/SIGCONT process freezes.

Operating Systems (CSCI 3310)

C / C++ · team of two

- Built a user-level thread library (ucontext context switching, FIFO scheduling, locks and condition variables under Mesa semantics, atomicity via interrupt masking) and an external virtual-memory pager (demand paging, second-chance clock replacement, software-tracked reference and dirty bits, disk swap), each validated with test suites written to expose buggy implementations.

EXPERIENCE

Software Engineering Intern, ProgressSoft Corporation

Amman, Jordan · Jun–Aug 2026

- Built EventGuard at a payments software company: a seven-module Java 21 and Spring Boot service that ingests payment files, validates records, detects duplicate payment IDs, and applies typed row-level rejection handling that separates parser errors from business-rule failures. github.com/adamzatar/EventGuard
- Implemented transactional persistence to PostgreSQL so a failed import cannot leave a partially written batch, with explicit commit and rollback and monetary values held as BigDecimal against a NUMERIC column to avoid floating-point drift.
- Kept a complete audit trail where rejected rows are preserved with a typed reason rather than discarded; verified with 107 passing tests including a live database integration test, working through merge requests and twice-weekly mentor code review.

SKILLS

Languages: Python, Java, TypeScript/JavaScript, SQL, C++, C

Backend & Web: FastAPI, Node.js, Fastify, Spring Boot, Spring MVC, REST API design, JDBC, transactional persistence, React, Next.js

Agentic AI: bounded tool-using agents, structured LLM outputs, evidence grounding and provenance, tool-boundary security over untrusted input, model/validator separation, human-in-the-loop review

Testing & Evaluation: test-driven development (pytest, JUnit 5, Mockito, Vitest), integration and concurrency testing, benchmarking, frozen holdout evaluation, code review, Git/GitLab, CI/CD

AI-Assisted Development: scaffolding, unfamiliar APIs, and test data; generated code validated against tests written first and reviewed line by line before it ships

Systems & Fundamentals: concurrency and synchronization, leases and fencing, crash recovery, memory management, distributed systems, data structures & algorithms, Linux/POSIX, Docker

Storage & Data: PostgreSQL, Redis, MySQL, SQLite, schema design, transactions

Research & Leadership: AI Layoffs and Investor Reactions (event study, regression analysis); Gibbons Research Fellowship (funded). Founder and President, Bowdoin Martial Arts Club; Student Advisor, CS Department faculty hiring

Spoken Languages: Arabic (native), English (fluent), German (intermediate)